

SOS Leh Analysis

Quarto

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Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

```
pre_workshop <- read.csv("Science of Sadharan (Responses).csv")
post_workshop <- read.csv("Science of Sadharan Form 2 (Responses).csv")

library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
pre_workshop <- pre_workshop %>%
  mutate(across(where(is.character), ~na_if(., "")))

post_workshop <- post_workshop %>%
  mutate(across(where(is.character), ~na_if(., "")))

career_dataset <- pre_workshop %>%
  rename(class = Grade.Class,
         school = School.Name,
         cq1 = I.would.like.to.have.a.career.in.1,
         cq2 = I.would.enjoy.a.career.in.1,
         cq3 = I.will.go.to.a.good.college.and.study.for.a.career.in.1,
         cq4 = I.will.have.a.successful.professional.career.and.make.substantial.scientific.
         cq5 = Some.day.when.I.tell.others.about.my.career..they.will.respect.me.for.doing.s
         cq6 = A.career.in.1.would.allow.me.to.work.with.others.in.meaningful.ways.) %>%
  select(Name, Age, Gender, class, school, cq1, cq2, cq3, cq4, cq5, cq6)

career_dataset$Gender <- as.factor(career_dataset$Gender)

career_dataset$cq1 <- reverse.code(keys = -1, items = career_dataset$cq1)
career_dataset$cq2 <- reverse.code(keys = -1, items = career_dataset$cq2)
```

```

career_dataset$cq3 <- reverse.code(keys = -1, items = career_dataset$cq3)
career_dataset$cq4 <- reverse.code(keys = -1, items = career_dataset$cq4)
career_dataset$cq5 <- reverse.code(keys = -1, items = career_dataset$cq5)
career_dataset$cq6 <- reverse.code(keys = -1, items = career_dataset$cq6)

```

```

career_dataset <- career_dataset %>%
  rowwise() %>%
  mutate(mean_score = mean(c_across(cq1:cq6), na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(group = "Pre") %>%
  mutate(row_num = row_number())

```

```

SC_dataset <- pre_workshop %>%
  rename(class = Grade.Class,
         school = School.Name,
         sc1 = My.family.has.encouraged.me.to.study.1,
         sc3 = Do.you.think.learning.1.is.important.for.your.daily.life.,
         sc4 = How.do.you.feel.when.you.learn.about.science.in.school.) %>%
  select(Name, Age, Gender, class, school, sc1, sc3, sc4)

SC_dataset$Gender <- as.factor(SC_dataset$Gender)

```

```

SC_dataset$sc4 <- recode(SC_dataset$sc4,
  "Bored" = 4,
  "Confused" = 3,
  "Curious" = 2,
  "Excited" = 1,
  "Good" = 1)
sum(is.na(SC_dataset$sc3))

```

```
[1] 124
```

```
sum(is.na(SC_dataset$sc4))
```

```
[1] 127
```

```
SC_dataset$sc1 <- reverse.code(keys = -1, items = SC_dataset$sc1)

SC_dataset$sc3 <- reverse.code(keys = -1, items = SC_dataset$sc3)

SC_dataset$sc4 <- reverse.code(keys = -1, items = SC_dataset$sc4)
```

```
SC_dataset <- SC_dataset %>%
  rowwise() %>%
  mutate(mean_score = mean(c_across(sc1:sc4), na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(group = "Pre") %>%
  mutate(row_num = row_number())

joined_pre <- full_join(career_dataset, SC_dataset, by = c("Name", "Age", "Gender", "class"),

joined_pre <- joined_pre %>%
  select(row_num, Name, Age, Gender, class, school, mean_score.x, mean_score.y, group) %>%
  rename(CI = mean_score.x,
         SCI = mean_score.y)
```

```
career_dataset2 <- post_workshop %>%
  rename(class = Grade.Class,
         school = School.Name.,
         CI = How.has.this.workshop.influenced.your.thoughts.on.pursuing.a.career.in.science)
  mutate(row_num = row_number()) %>%
  select(row_num, Name, Age, Gender, class, school, CI) %>%
  mutate(group = "Post")

career_dataset2$Gender <- as.factor(career_dataset2$Gender)
career_dataset2$CI <- reverse.code(keys = -1, items = career_dataset2$CI)
```

```
sc_dataset2 <- post_workshop %>%
  rename(class = Grade.Class,
         school = School.Name.,
         s1 = How.often.will.you.think.about.science.now.after.this.workshop.,
         s2 = Did.the.workshop.make.you.more.interested.in.learning.about.how.things.work.) %>%
  mutate(row_num = row_number()) %>%
  select(row_num, Name, Age, Gender, class, school, s1, s2)

sc_dataset2$Gender <- as.factor(sc_dataset2$Gender)
```

```

sc_dataset2$s1[sc_dataset2$s1 == 5] <- 4

sc_dataset2$s1 <- reverse.code(keys = -1, items = sc_dataset2$s1)
sc_dataset2$s2 <- reverse.code(keys = -1, items = sc_dataset2$s2)

sc_dataset2 <- sc_dataset2 %>%
  rowwise() %>%
  mutate(SCI = mean(c_across(s1:s2), na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(group = "Post") %>%
  mutate(row_num = row_number())

joined_post <- full_join(career_dataset2, sc_dataset2, by = c("Name", "Age", "Gender", "class"))

joined_post <- joined_post %>%
  select(row_num, Name, Age, Gender, class, school, CI, SCI, group)

combined <- bind_rows(joined_pre, joined_post)
combined$group <- factor(combined$group, levels = c("Pre", "Post"))

```

```
sum(is.na(combined$CI))
```

```
[1] 3
```

```
sum(is.na(combined$SCI))
```

```
[1] 1
```

```

mean_sd <- combined %>%
  group_by(group) %>%
  summarise(mean_SCI = mean(SCI, na.rm = T),
            mean_CI = mean(CI, na.rm = T),
            SD_SCI = sd(SCI, na.rm = T),
            SD_CI = sd(CI, na.rm = T))
mean_sd

```

```

# A tibble: 2 x 5
  group mean_SCI mean_CI SD_SCI SD_CI
  <fct>   <dbl>   <dbl> <dbl> <dbl>
1 Pre     2.79     3.06  1.02  0.648
2 Post     3.34     3.35  0.626  0.742

```

```
shapiro.test(combined$CI)
```

Shapiro-Wilk normality test

```
data: combined$CI  
W = 0.88581, p-value < 2.2e-16
```

```
shapiro.test(combined$SCI)
```

Shapiro-Wilk normality test

```
data: combined$SCI  
W = 0.86747, p-value < 2.2e-16
```

```
write.csv(combined, "my_data.csv")
```

```
combined %>%  
  group_by(class) %>%  
  summarise(n())
```

```
# A tibble: 15 x 2  
  class  `n()`  
  <chr>  <int>  
1 "10"      6  
2 "10th"    68  
3 "6"      15  
4 "6th"     27  
5 "6th "    7  
6 "7"      32  
7 "7TH"     2  
8 "7th"     73  
9 "7th "    20  
10 "8"       1  
11 "8th"    118  
12 "8th "    8  
13 "9"      15  
14 "9th"     95  
15 "9th "    2
```

```
combined$class <- as.numeric(gsub("\\D", "", combined$class))
```

```
combined %>%
  group_by(class) %>%
  summarise(n())
```

```
# A tibble: 5 x 2
```

```
  class `n()`
  <dbl> <int>
1     6    49
2     7   127
3     8   127
4     9   112
5    10    74
```

```
combined$school[combined$school == 12] <- NA
combined$school[combined$school %in% c("G.M.H.S. Lankore",
                                       "GMHS Lankore",
                                       "GMHS Lankore, Kargi;",
                                       "PM Shiri GMHS ")] <- "GMHS"
```

```
combined$school[combined$school == "GMHS Lankore "] <- "GMHS"
```

```
combined %>%
  group_by(school) %>%
  summarise(n())
```

```
# A tibble: 25 x 2
```

```
  school      `n()`
  <chr>      <int>
1 "GMHS"      50
2 "IMS"       42
3 "IMS "       8
4 "IMS Chochut" 1
5 "Imamiya"    20
6 "Imamiya Mission School" 2
7 "Imamiya Public School" 1
8 "Ims"        1
9 "Jamyang"     3
10 "Jamyang "  45
```

```
# i 15 more rows
```

```

combined$school[combined$school %in% c("IMS ",
                                         "IMS Chochut",
                                         "Imamiya",
                                         "Imamiya Mission School",
                                         "Ims",
                                         "Smamiya Mission School",
                                         "Imamiya Public School")] <- "IMS"

combined$school[combined$school %in% c("Jamyang ",
                                         "Jamyang",
                                         "Jamyang High School",
                                         "Jamyang High School ",
                                         "Jamyang High School - Leh",
                                         "Jamyang High school",
                                         "Jamyang School",
                                         "Jamyang School, Leh",
                                         "Jamyang High School",
                                         "Jamyang High school ")] <- "JHS"
combined$school[combined$school == "Jamyang High School "] <- "JHS"

combined$school[combined$school %in% c("NMCE",
                                         "Millenium",
                                         "NMCS ",
                                         "New Millennium ",
                                         "Ims")] <- "NMCS"

combined %>%
  group_by(school) %>%
  summarise(n())

```

```

# A tibble: 6 x 2
  school `n()`
  <chr>   <int>
1 GMHS      50
2 IMS       76
3 JHS      108
4 NMCS     216
5 SB         1
6 <NA>      38

```



```
combined$school[combined$school == "SB"] <- "IMS"
```

```
write.csv(combined, "my_data_final.csv")
```

```
combined$Age <- as.factor(combined$Age)
combined$class <- as.factor(combined$class)
combined$school <- as.factor(combined$school)
combined$CI <- as.numeric(combined$CI)
combined$SCI <- as.numeric(combined$SCI)
glimpse(combined)
```

Rows: 489

Columns: 9

```
$ row_num <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, ~
$ Name      <chr> "Mohd. Hassnain Ali", "Asiya Batlool", "Shahida Kousar", "Sheh~
$ Age       <fct> 15, 14, 14, 14, 16, 17, 17, 15, 14, 16, 16, 14, 14, 14, 15, 16~
$ Gender    <fct> Male, Female, Female, Female, Female, Female, Female, Female, ~
$ class     <fct> 8, 9, 8, 8, 10, 10, 10, 9, 9, 9, 10, 8, 8, 9, 10, 9, 10, 9, 10~
$ school    <fct> GMHS, GMHS, GMHS, GMHS, GMHS, GMHS, GMHS, GMHS, GMHS, GMHS, GM~
$ CI        <dbl> 1.666667, 2.500000, 3.666667, 3.833333, 3.666667, 3.833333, 3.~
$ SCI       <dbl> 1, 3, 4, 4, 4, 2, 2, 3, 4, 4, 4, 3, 4, 1, 2, 3, 4, 4, 4, 4, ~
$ group     <fct> Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pre, Pr~
```

```
combined %>%
  group_by(group, Gender) %>%
  summarise(Mean_SCI = mean(SCI, na.rm = T),
            median_SCI = median(SCI, na.rm = T),
            IQR_SCI = IQR(SCI, na.rm = T),
            Mean_CI = mean(CI, na.rm = T),
            median_CI = median(CI, na.rm = T),
            IQR_CI = IQR(CI, na.rm = T))
```

`summarise()` has grouped output by 'group'. You can override using the
`.groups` argument.

A tibble: 6 x 8

Groups: group [2]

	group	Gender	Mean_SCI	median_SCI	IQR_SCI	Mean_CI	median_CI	IQR_CI
	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Pre	Female	2.87	3	1.67	3.07	3.17	1.12

2	Pre	Male	2.63	2.83	1.33	3.03	3.17	0.758
3	Pre	<NA>	3	3	0	3	3	0
4	Post	Female	3.33	3.5	1	3.36	3	1
5	Post	Male	3.34	3.5	1	3.33	4	1
6	Post	<NA>	3	3	0	3	3	0

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.1      v stringr 1.5.2
v lubridate 1.9.4    v tibble 3.3.0
v purrr 1.1.0      v tidyr 1.3.1
v readr 2.1.5
-- Conflicts ----- tidyverse_conflicts() --
x psych::%+%( ) masks ggplot2::%+%( )
x psych::alpha( ) masks ggplot2::alpha( )
x dplyr::filter( ) masks stats::filter( )
x dplyr::lag( ) masks stats::lag( )
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ARTool)
```

```
kruskal.test(SCI ~ interaction(group, school), data = combined)
```

Kruskal-Wallis rank sum test

```
data: SCI by interaction(group, school)
Kruskal-Wallis chi-squared = 80.827, df = 7, p-value = 9.342e-15
```

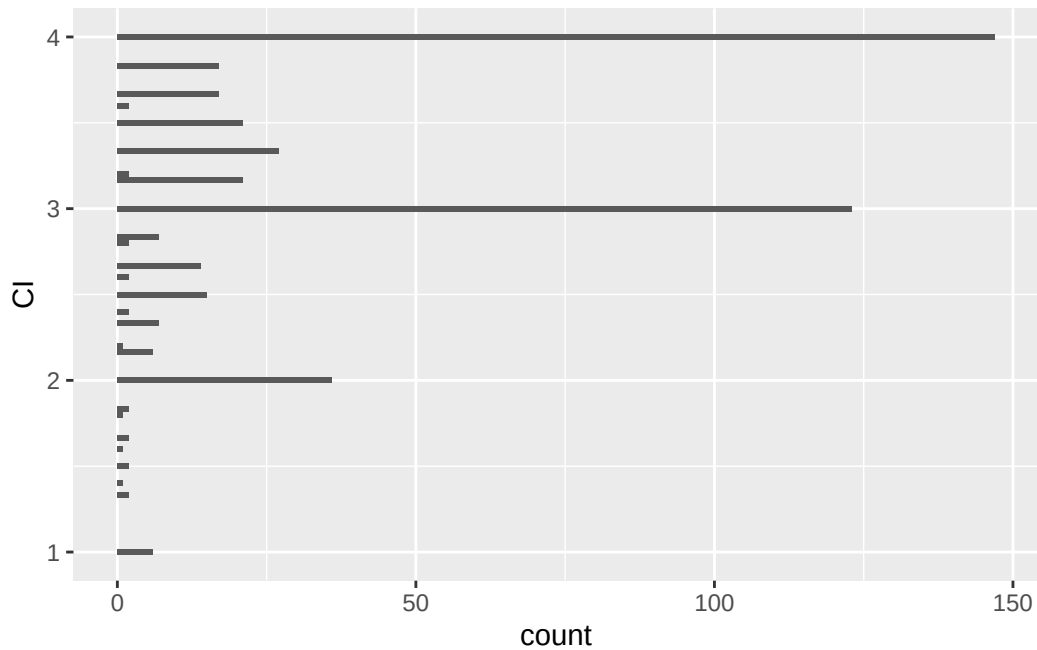
```
kruskal.test(CI ~ interaction(group, school), data = combined)
```

Kruskal-Wallis rank sum test

```
data: CI by interaction(group, school)
Kruskal-Wallis chi-squared = 59.101, df = 7, p-value = 2.282e-10
```

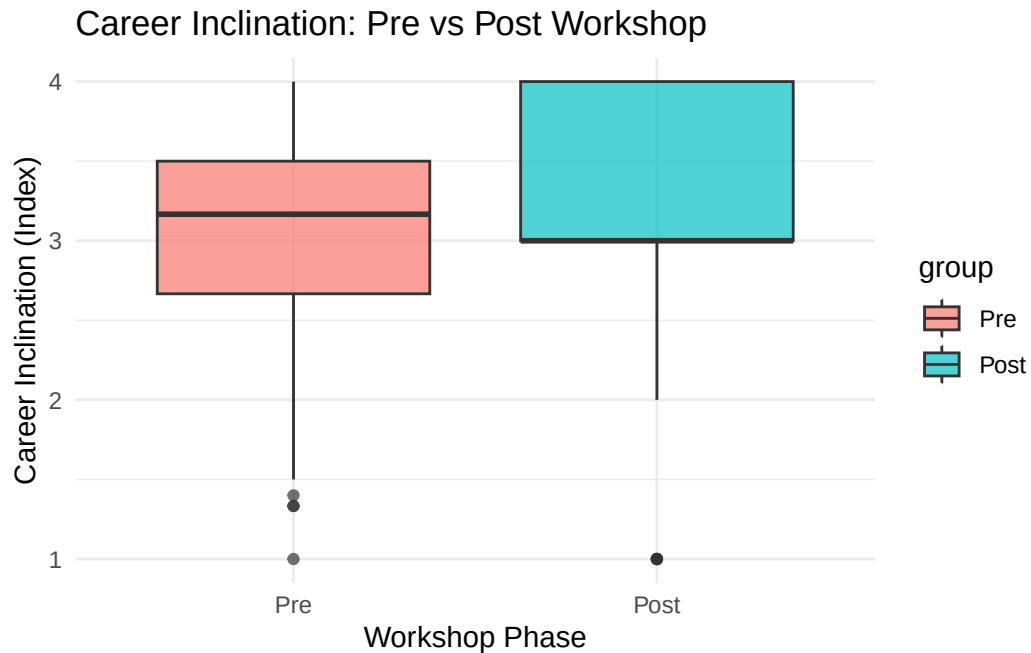
```
ggplot(data = combined) + geom_bar(mapping = aes(y = CI, group = group))
```

Warning: Removed 3 rows containing non-finite outside the scale range (``stat_count()``).



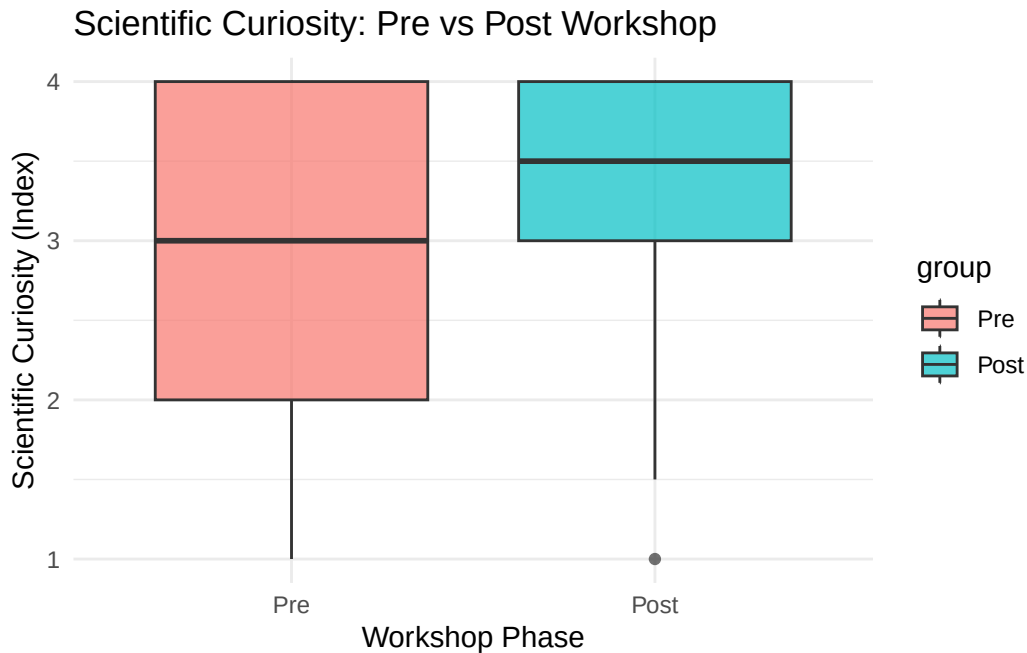
```
ggplot(combined, aes(x = group, y = CI, fill = group)) +  
  geom_boxplot(alpha = 0.7) +  
  labs(title = "Career Inclination: Pre vs Post Workshop",  
        x = "Workshop Phase", y = "Career Inclination (Index)") +  
  theme_minimal()
```

Warning: Removed 3 rows containing non-finite outside the scale range (``stat_boxplot()``).



```
# Violin plot
ggplot(combined, aes(x = group, y = SCI, fill = group)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Scientific Curiosity: Pre vs Post Workshop",
       x = "Workshop Phase", y = "Scientific Curiosity (Index)") +
  theme_minimal()
```

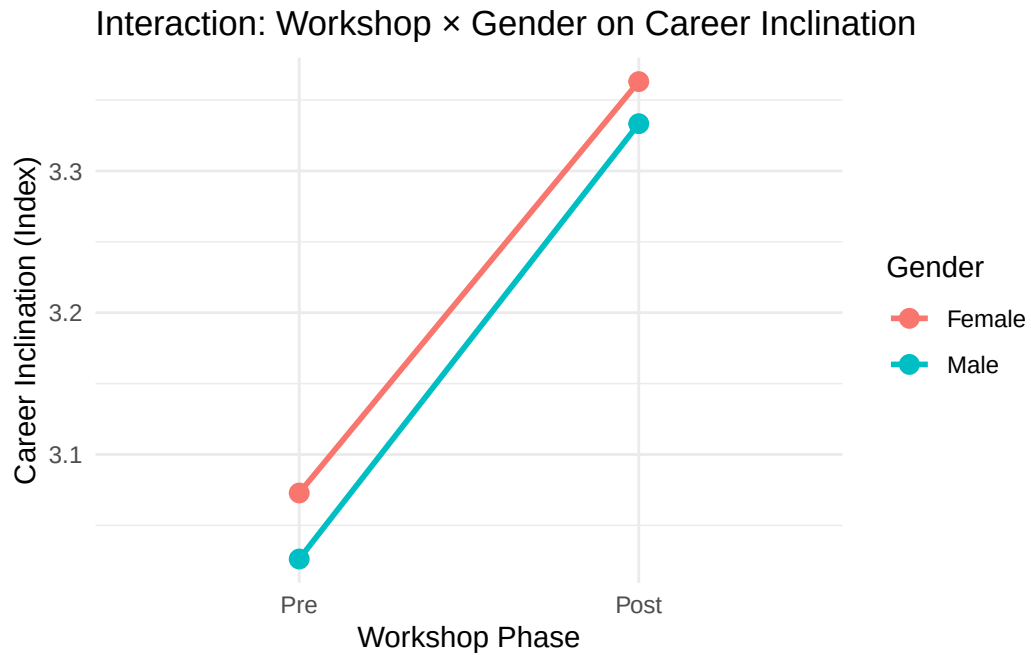
Warning: Removed 1 row containing non-finite outside the scale range (``stat_boxplot()``).



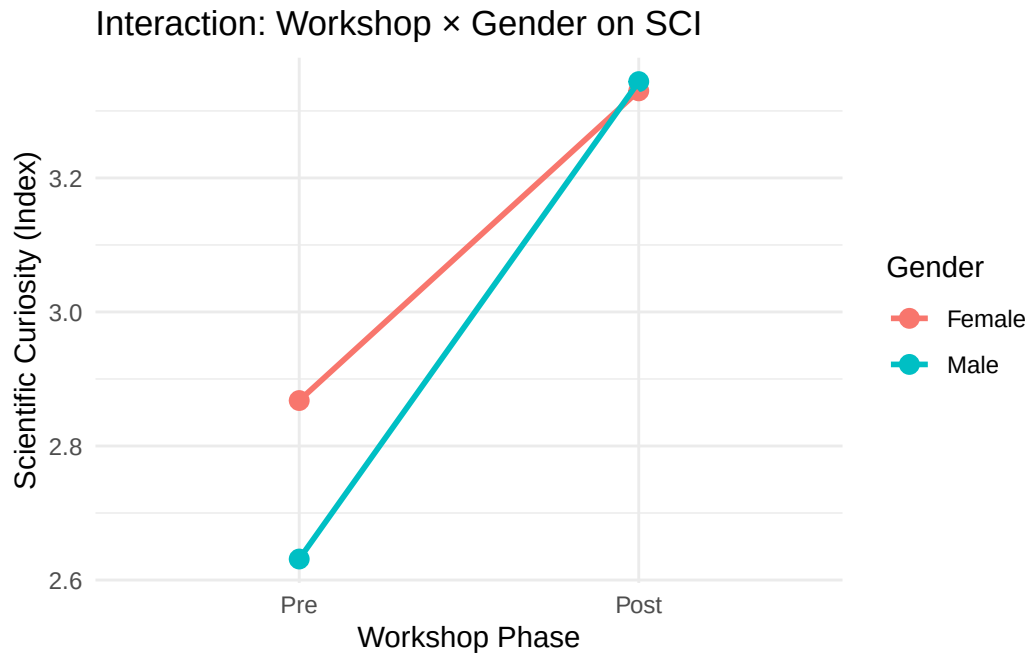
```
summary <- combined %>%
  group_by(group, Gender) %>%
  summarise(mean_CI = mean(CI, na.rm = T),
            sd_CI = sd(CI, na.rm = T),
            mean_SCI = mean(SCI, na.rm = T),
            sd_SCI = sd(SCI, na.rm = T)) %>%
  filter(!is.na(Gender))
```

`summarise()` has grouped output by 'group'. You can override using the `groups` argument.

```
ggplot(summary, aes(x = group, y = mean_CI, color = Gender, group = Gender)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(title = "Interaction: Workshop × Gender on Career Inclination",
       x = "Workshop Phase", y = "Career Inclination (Index)") +
  theme_minimal()
```



```
ggplot(summary, aes(x = group, y = mean_SCI, color = Gender, group = Gender)) +  
  geom_point(size = 3) +  
  geom_line(linewidth = 1) +  
  labs(title = "Interaction: Workshop × Gender on SCI",  
        x = "Workshop Phase", y = "Scientific Curiosity (Index)") +  
  theme_minimal()
```



```
combined_new <- combined %>%  
  filter(!is.na(Gender), !is.na(CI), !is.na(SCI))  
  
art_model <- art(SCI ~ group * Gender, data = combined_new)
```